

Chapter 1: Introduction

Our goal:

- ❑ get “feel” and terminology
- ❑ more depth, detail *later* in course
- ❑ approach:
 - ❖ use Internet as example

Overview:

- ❑ what's the Internet?
- ❑ what's a protocol?
- ❑ network edge: hosts, access nets, physical media
- ❑ network core: packet/circuit switching, Internet structure
- ❑ performance: loss, delay, throughput
- ❑ protocol layers, service models
- ❑ security
- ❑ history

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Chapter 1: roadmap

1.1 What *is* the Internet?

1.2 Network edge

- ❑ end systems, access networks, links

1.3 Network core

- ❑ circuit switching, packet switching, network structure

1.4 Delay, loss and throughput in packet-switched networks

1.5 Protocol layers, service models

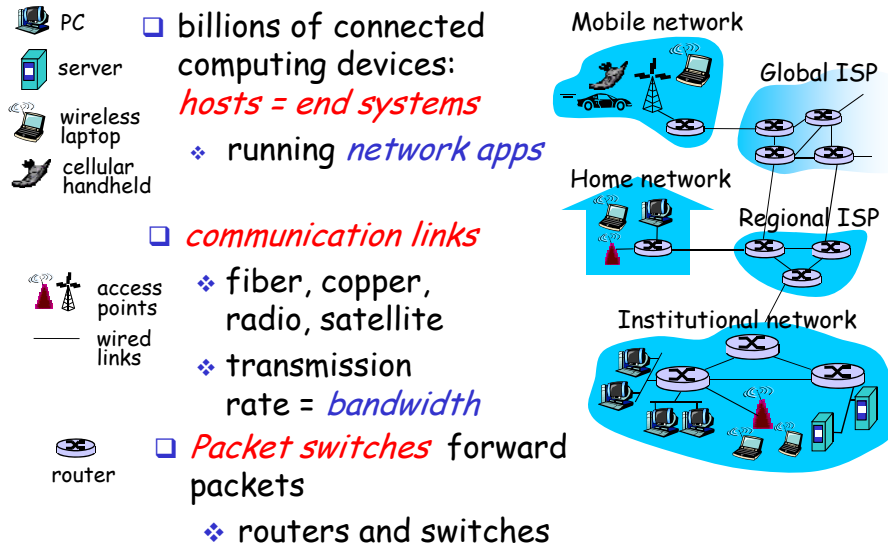
1.6 Networks under attack: security

1.7 History

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What's the Internet: "nuts and bolts" view

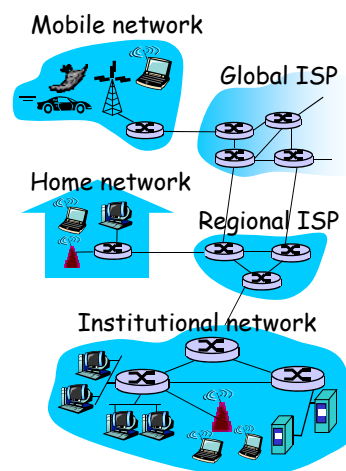


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What's the Internet: architecture & protocols

- Internet: "network of networks"*
 - loosely hierarchical
 - public Internet versus private intranet
- protocols* control sending, receiving of msgs
 - e.g., TCP, IP, HTTP, Skype, Ethernet
- Internet standards
 - RFC: Request for comments
 - IETF: Internet Engineering Task Force

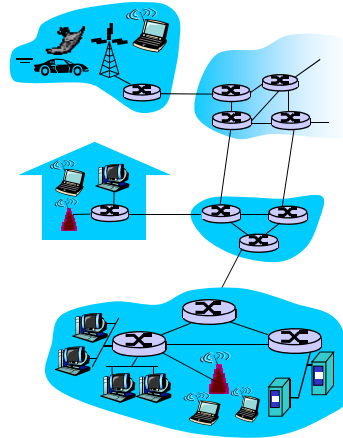


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What's the Internet: a service view

- **communication infrastructure** enables distributed applications:
 - ❖ Web, VoIP, email, games, e-commerce, file sharing
- **communication services provided to apps:**
 - ❖ reliable data delivery from source to destination
 - ❖ "best effort" (unreliable) data delivery



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What's a protocol?

human protocols:

- "what's the time?"
- "I have a question"
- introductions

network protocols:

- machines rather than humans
- all communication activity in Internet governed by protocols

protocols define format, order of msgs sent and received among network entities, and actions taken on msg transmission, receipt, or timeout

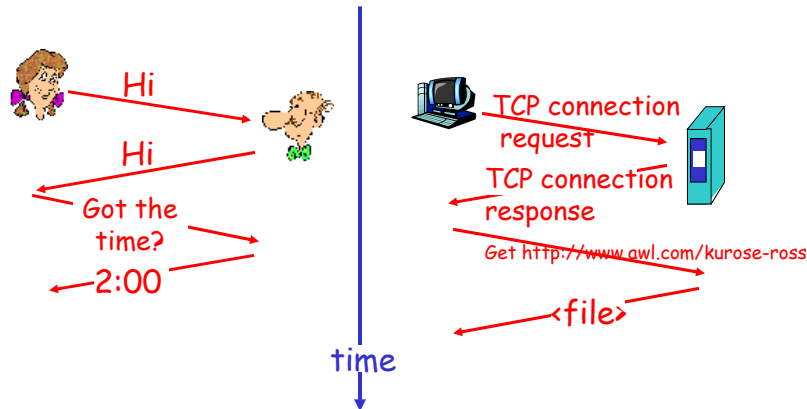
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What's a protocol?

a human protocol and a computer network protocol:



Q: Other human protocols?

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Chapter 1: roadmap

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□ end systems, access networks, links

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□ circuit switching, packet switching, network structure

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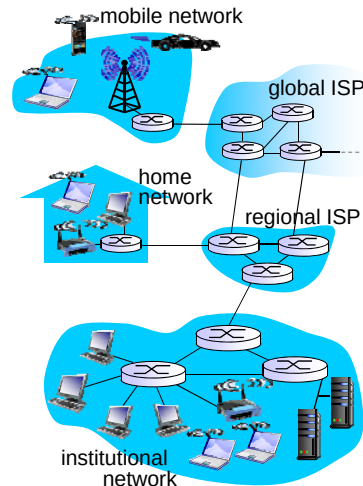
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A closer look at network structure:

- ❑ *network edge:*
 - ❖ hosts: clients and servers
 - ❖ servers often in data centers
- ❑ *access networks, physical media:* wired, wireless communication links
- ❑ *network core:*
 - ❖ interconnected routers
 - ❖ network of networks



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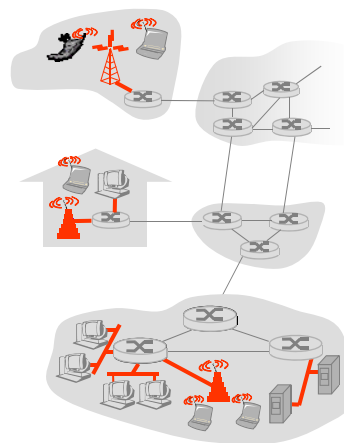
Access networks and physical media

Q: How to connect end systems to edge router?

- ❑ residential access nets
- ❑ institutional access networks (school, company)
- ❑ mobile access networks

Keep in mind:

- ❑ bandwidth (bits per second) of access network?
- ❑ shared or dedicated?



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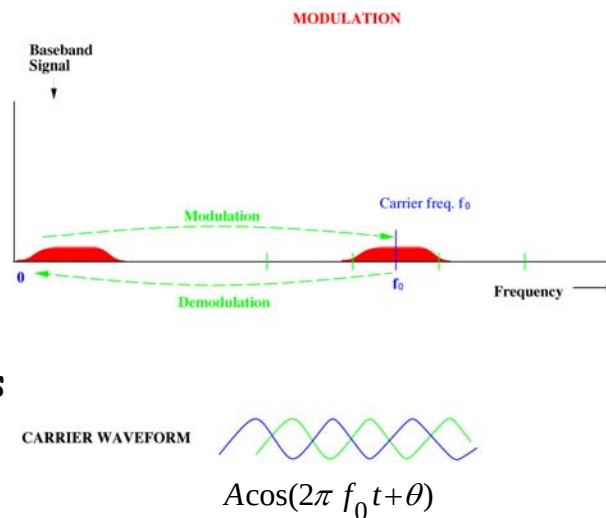
From physical media to communication channels—basic concepts

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Modulation and Demodulation

- Common examples: radio, television channels for **analog** signals
- Can also be used for **digital** signals (encoding binary data)



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Shannon's Theorem

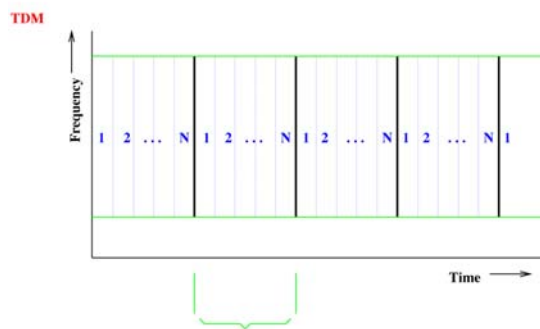
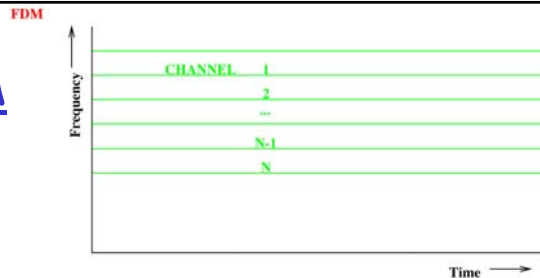
$$C = B \log_2 (1 + S/N)$$

where C max capacity in bits/sec
 B bandwidth in hertz
 S/N signal to noise ratio

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FDM vs. TDM



Duration of frame (or superframe) is 125 μ sec in digital telephone networks

Duration of frame or superframe (125 μ sec in digital telephone network)

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TDM in Telephone Networks

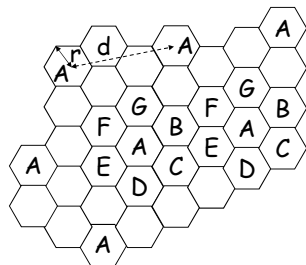
- ❑ Why 125 μsec for frame duration?
- ❑ **Sampling Theorem:**
An analog signal can be reconstructed from samples taken at a rate equal to twice the signal bandwidth
- ❑ Bandwidth for voice signals is 4 KHz; for hi fidelity music, 22.05 KHz
- ❑ Sampling rate for voice = 8000 samples/sec or one voice sample every 125 μsec
- ❑ Digital voice channel (uncompressed),
8 bits $\times$ 8000/sec = 64 Kbps

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Other Multiplexing Techniques

- ❑ Space division multiplex
 - ❖ Same frequency used in different cables
 - ❖ Same frequency used in different (nonadjacent) cells
- ❑ Wavelength division multiplex
 - ❖ Light pulses sent at different wavelengths in optical fiber
- ❑ Code division multiplex (in chapter 6 of text)



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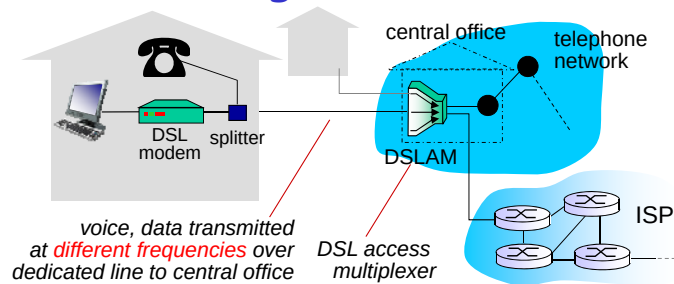
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Now back to access networks

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Access net: digital subscriber line (DSL)

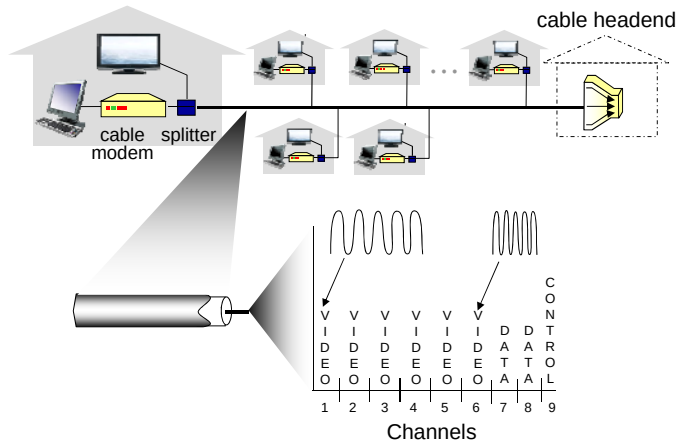


- ❖ use FDM in telephone line to central office DSLAM
 - data over DSL line goes to Internet
 - voice over DSL line goes to telephone net
- ❖ < 2.5 Mbps upstream transmission rate (typically < 1 Mbps)
- ❖ < 24 Mbps downstream transmission rate (typically < 10 Mbps)

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Access net: cable network

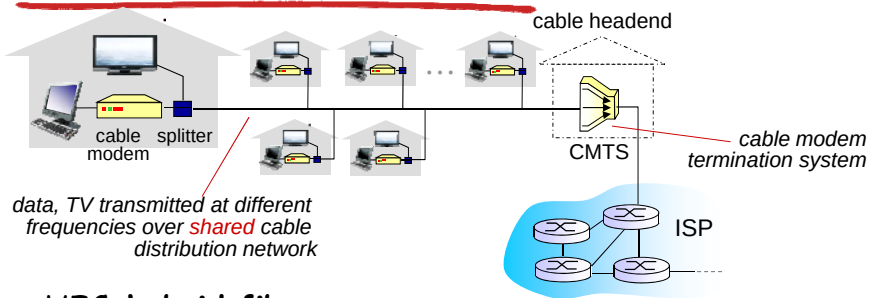


frequency division multiplexing: different channels transmitted in different frequency bands

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Access net: cable and fiber



❖ HFC: hybrid fiber coax

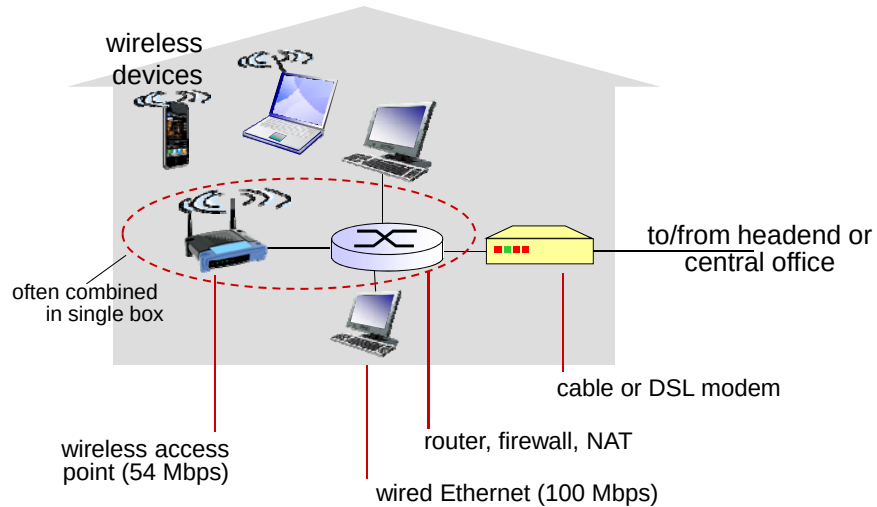
- cable and fiber network attaches homes to ISP router
- asymmetric: up to 30 Mbps downstream transmission rate, 2 Mbps upstream transmission rate
- homes *share access network* to cable headend (unlike DSL, which has dedicated access to central office)

❖ Fiber to the home (Verizon, Google) - optical switches

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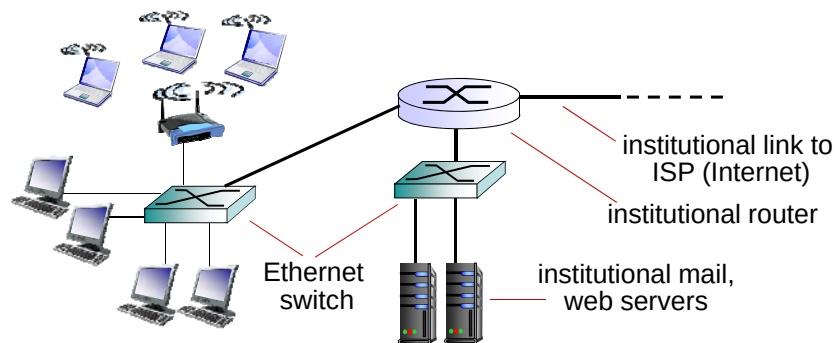
Access net: home network



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Enterprise access networks (Ethernet)



- ❖ today, end systems typically connect into Ethernet switch
 - 10 Mbps, 100Mbps, 1Gbps, 10Gbps transmission rates
- ❖ A large enterprise network is connected to multiple ISPs

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Wireless access networks

- shared *wireless* access network connects end system to router
 - ❖ via base station aka “access point”

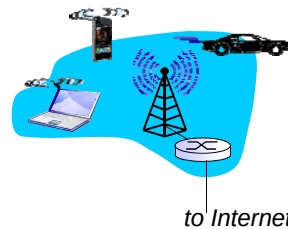
wireless LANs:

- within building (100 ft)
- 802.11b/g/n (WiFi)



wide-area wireless access

- provided by telco (cellular) operator, 10's km
- between 1 and 10 Mbps
- 3G, 4G: LTE



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Physical Media

- **Bit:** propagates between transmitter & receiver
- **physical link:** what lies between transmitter & receiver
- **guided media:**
 - ❖ signals propagate inside solid media: copper, fiber, coax
- **unguided media:**
 - ❖ signals propagate freely, e.g., radio

Twisted Pair (TP)

- two insulated copper wires
 - ❖ Category 3: traditional phone wires, 10 Mbps Ethernet
 - ❖ Category 5: 100Mbps Ethernet
 - ❖ Category 6: 10Gbps



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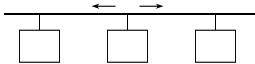
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Physical Media: coax, fiber



Coaxial cable:

- ❑ two concentric copper conductors
 - ❑ baseband:
 - ❖ single channel in cable
 - ❖ legacy Ethernet
 - ❑ broadband:
 - ❖ multiple channels in cable
 - ❖ HFC
 - ❑ bidirectional
- *broadcast*



Fiber optic cable:

- ❑ glass fiber carrying light pulses
- ❑ high-speed point-to-point transmission, e.g., 10's-100's Gps
- ❑ low error rate:
 - ❖ repeaters spaced far apart
 - ❖ immune to electromagnetic noise

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Physical media: radio

- ❑ signal carried in electromagnetic waves
- ❑ can be *omnidirectional* or as a *directional beam*
- ❑ propagation environment effects:
 - ❖ reflection
 - ❖ obstruction by objects
 - ❖ interference

Radio link types:

- ❑ LAN (e.g., Wi Fi)
 - ❖ 11Mbps, 54 Mbps
- ❑ wide-area (e.g., cellular)
 - ❖ 3G cellular: ~ a few Mbps
- ❑ terrestrial microwave
 - ❖ e.g. up to 45 Mbps channels
- ❑ satellite
 - ❖ 45Mbps channel (or multiple smaller channels)
 - ❖ *geosynchronous* vs. *low altitude*
 - ❖ 270 msec end-end delay for geosynchronous

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Chapter 1: roadmap

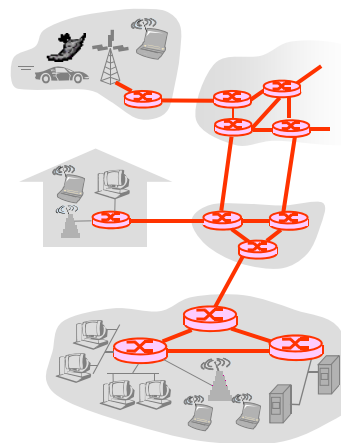
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 - circuit switching, packet switching, network structure
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The Network Core

- mesh of interconnected routers
- the fundamental question: how is data transferred through net?
 - ❖ circuit switching: dedicated circuit per call: telephone net
 - ❖ packet-switching: data sent thru net in discrete "chunks"



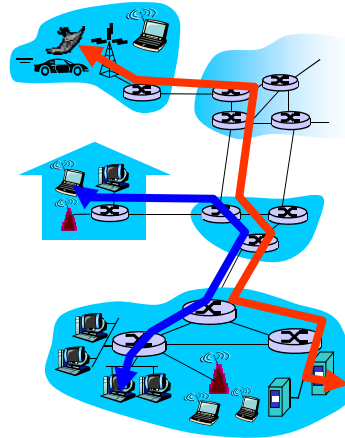
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Network Core: Circuit Switching

End-to-end resources reserved for each "call"

- ❑ E.g., link bandwidth
 - ❖ FDM, TDM
- ❑ end-to-end circuit-like (guaranteed) performance
- ❑ call setup required
 - ❖ resource piece *idle* if not used by the call (*no sharing*)



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Numerical example

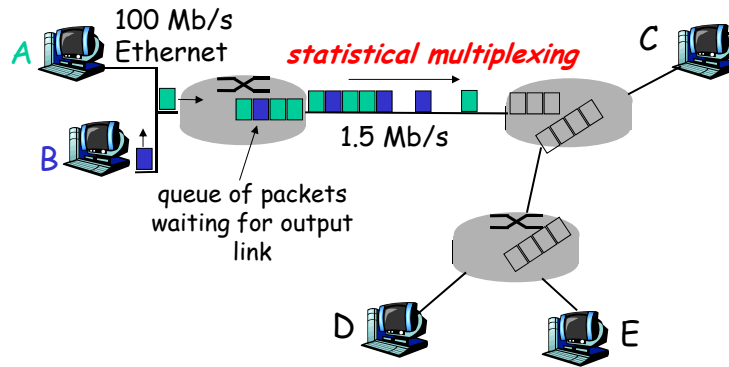
- ❑ How long does it take to send a file of 640,000 bits from host A to host B over a circuit-switched network?
 - ❖ all links are 1.536 Mbps
 - ❖ each link uses TDM with 24 slots/sec (i.e., one slot per circuit)
 - ❖ 500 msec to establish end-to-end circuit

Let's work it out!

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Packet Switching: Statistical Multiplexing



- ❑ Sequence of A & B packets does not have fixed pattern
bandwidth shared on demand → *statistical multiplexing*
- ❑ queueing delay, packet loss
- ❑ also called asynchronous time division multiplexing (ATDM)

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Network Core: Packet Switching

*each end-end data stream
divided into packets*

- ❑ packets of different users
share network resources
- ❑ each packet uses full link
bandwidth
- ❑ resources used *as needed*

Bandwidth division into "pieces"
Dedicated allocation
Resource reservation

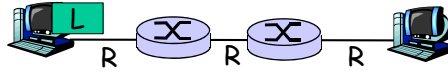
resource contention:

- ❑ aggregate resource
demand can exceed
amount available
- ❑ **congestion**: packets
queue, wait for link use
- ❑ **store and forward**:
packets move one hop
at a time
 - ❖ Each node receives the
complete packet before
forwarding

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Disadvantage of store-and-forward



- ❑ takes L/R seconds to transmit (push out) a message of L bits on to link at R bps

Example:

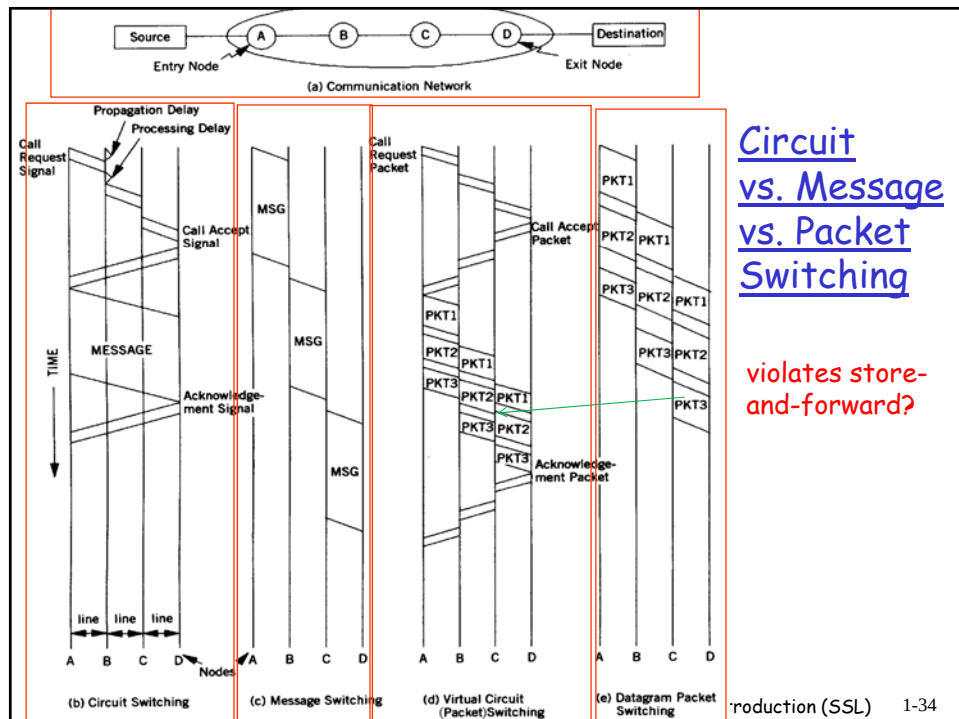
- ❖ $L = 7.5$ Mbits
- ❖ $R = 1.5$ Mbps
- ❖ End-to-end delay more than 15 seconds

- ❑ *store and forward:* entire message must arrive at router before it can be transmitted on next link

- ❑ A file/message larger than **maximum packet size** is transmitted as multiple packets

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Packet Switching versus Message Switching

Advantages of packet switching

- ❑ Smaller end-to-end delay from pipelining
- ❑ Less data loss from transmission errors

Disadvantages of packet switching

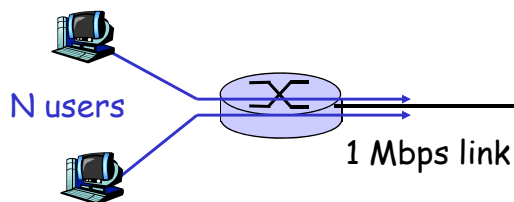
- ❑ More header bits
- ❑ Additional work to do segmentation and reassembly

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Packet switching versus circuit switching

- ❑ 1 Mb/s link
- ❑ each user:
 - ❖ 100 kb/s when "active"
 - ❖ active 10% of time (a "bursty" user)
- ❑ *circuit-switching*:
 - ❖ 10 users
- ❑ *packet switching*:
 - ❖ with 35 users, probability > 10 active at same time is less than .0004



Q: how did we get value 0.0004?

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Packet switching versus circuit switching

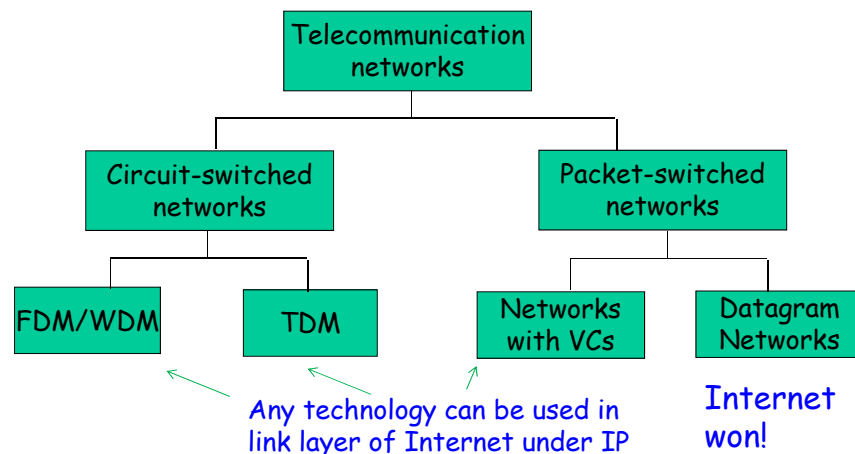
Is packet switching a "slam dunk winner?"

- ❑ great for bursty data
 - ❖ resource sharing
 - ❖ simpler, no call setup
- ❑ excessive congestion → packet delay and loss
 - ❖ protocols needed for reliable data transfer, congestion control
- ❑ Q: How to provide circuit-like behavior?
 - ❖ bandwidth guarantees needed for *interactive* audio/video apps
 - ❖ solution may impact network neutrality

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Network Taxonomy



Aside - Internet's transport layer provides both connection-oriented (TCP) and connectionless services (UDP).

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Internet structure: network of networks

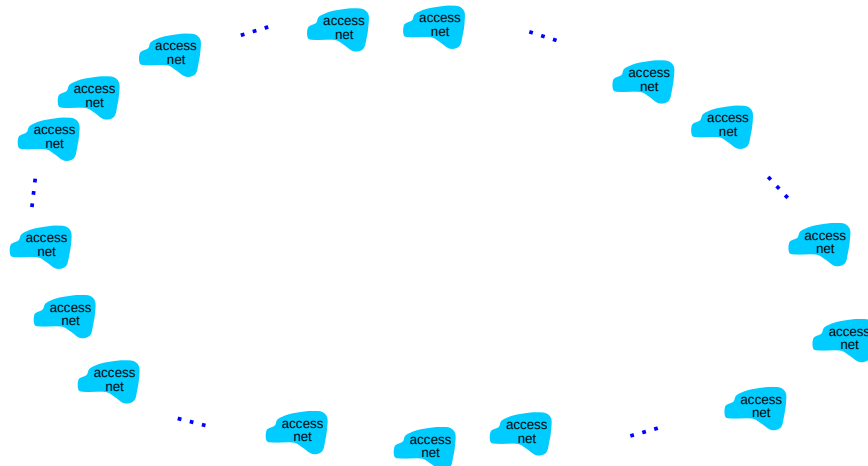
- ❑ End systems connect to Internet via **access ISPs** (Internet Service Providers)
 - ❖ Residential, company, and university ISPs
- ❑ Access ISPs in turn must be interconnected
 - ❖ so that any two hosts can send packets to each other
- ❑ Resulting network of networks is very complex
 - ❖ Evolution was driven by economics and national policies

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Internet structure: network of networks

Question: given *millions* of access ISPs, how to connect them together?

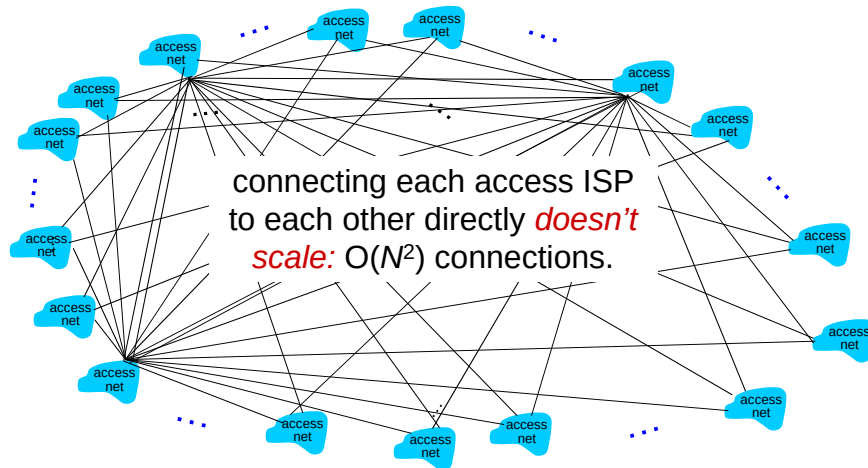


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Internet structure: network of networks

Option: connect each access ISP to every other access ISP?

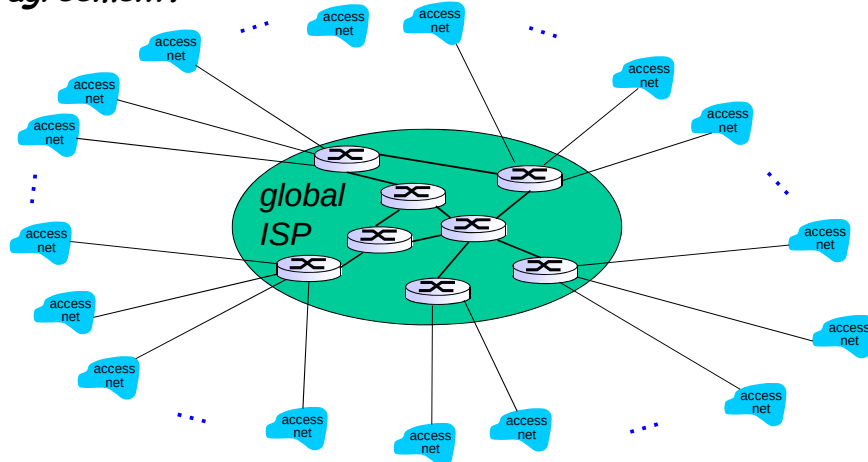


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Internet structure: network of networks

Option: connect each access ISP to a global transit ISP? Customer and provider ISPs have economic agreement.

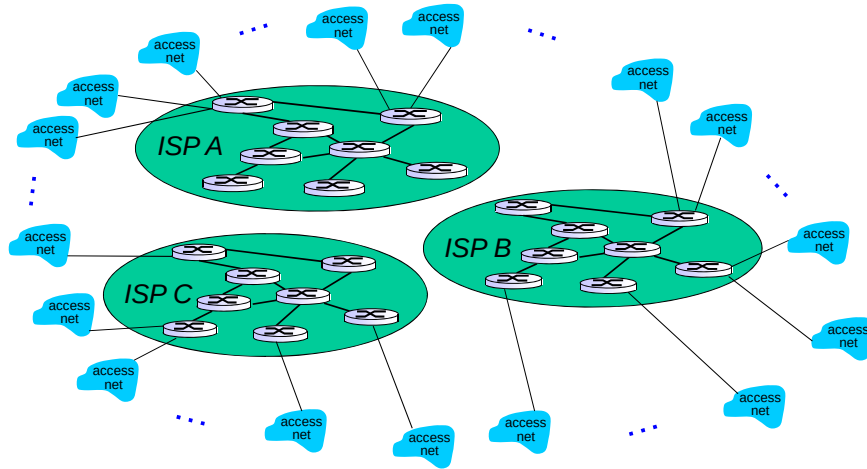


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Internet structure: network of networks

But if one global ISP is viable business, there will be competitors

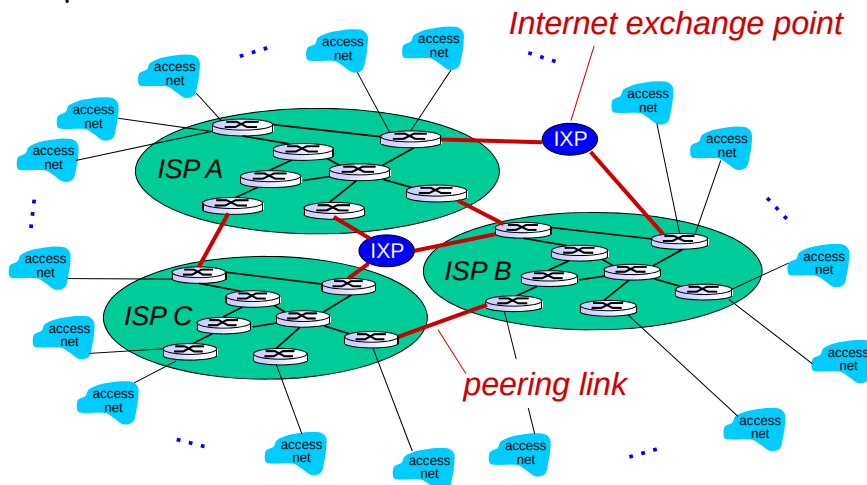


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Internet structure: network of networks

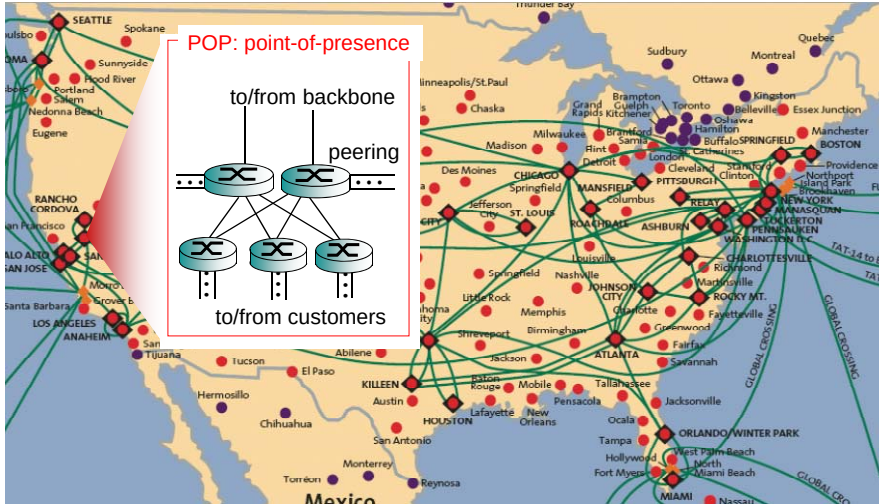
But if one global ISP is viable business, there will be competitors which must be interconnected



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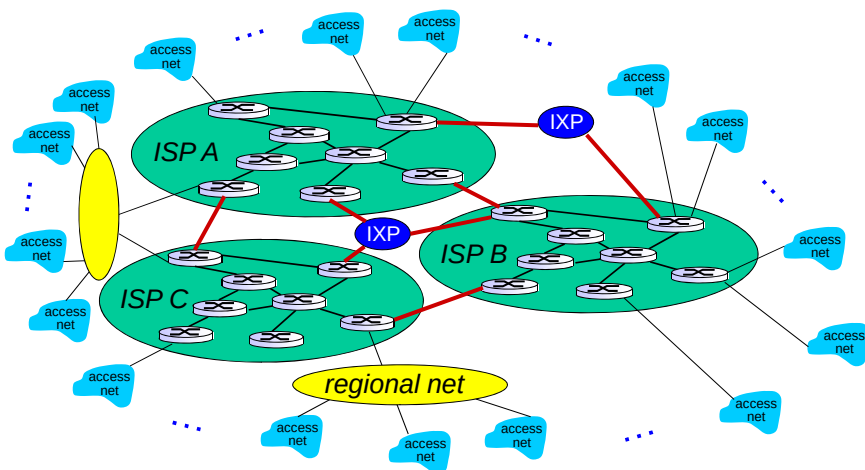
Tier-1 ISP: e.g., Sprint



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Internet structure: network of networks

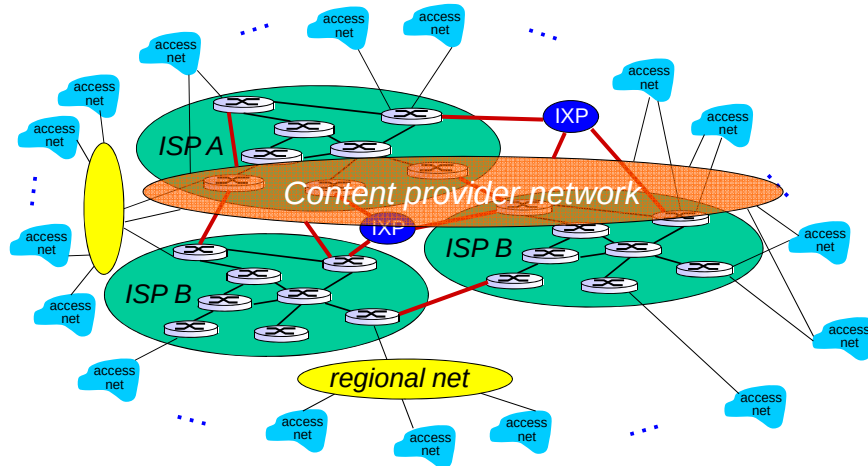
... and regional networks may arise to connect access nets to ISPS



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Internet structure: network of networks

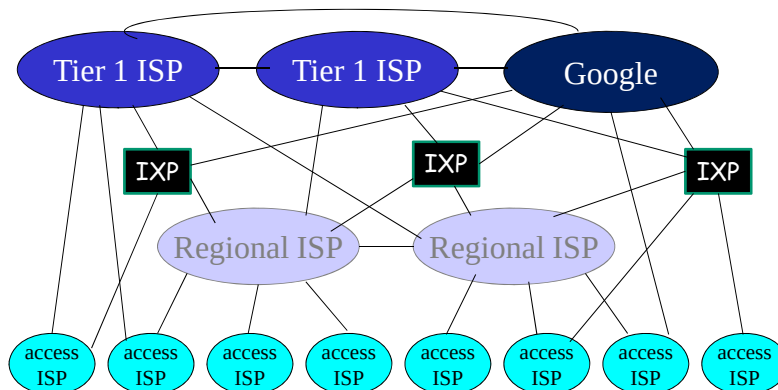
... and a content provider network (e.g., Akamai, Google, Microsoft) may run its own network to bring services, content close to end users



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Internet structure: network of networks



- at center: small # of well-connected large networks
 - ❖ “tier-1” commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
 - ❖ content provider networks (e.g., Google): private network that connects its data centers to Internet, often bypassing tier-1 and regional ISPs

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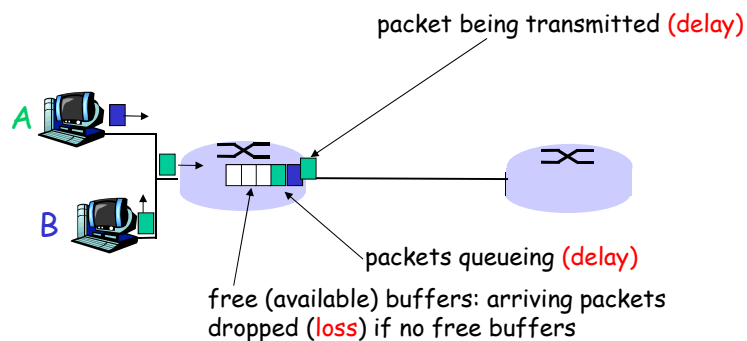
1.7 History

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How do loss and delay occur?

- packet arrival rate to link temporarily exceeds output link capacity
- packets queue, wait for turn



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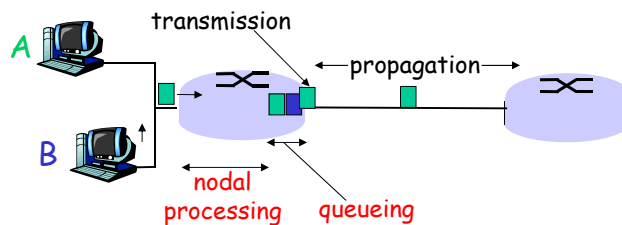
Four sources of packet delay

1. nodal processing:

- ❖ check bit errors
- ❖ determine output link

2. queueing

- ❖ time waiting at output link for transmission
- ❖ depends on congestion level of router



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Delay in packet-switched networks

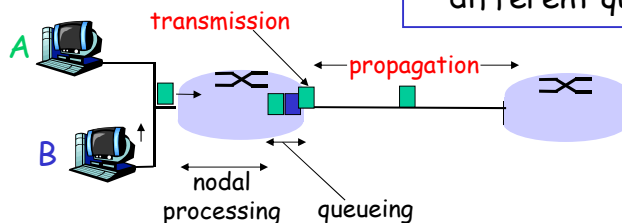
3. Transmission delay:

- ❑ R : link bandwidth (bps)
- ❑ L : packet length (bits)
- ❑ time to send bits into link = L/R

4. Propagation delay:

- ❑ d : length of physical link
- ❑ s : propagation speed in medium ($\sim 2 \times 10^8$ m/sec)
- ❑ propagation delay = d/s

Note: s and R are very different quantities!



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End-to-End Delay

- Nodal delay (from when last bit of packet arrives at this node to when last bit arrives at next node)

$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

- End-to-end delay over N identical nodes/links from client c to server s (from when last bit of packet leaves client to when last bit arrives at server)

$$d_{c-s} = d_{prop} + Nd_{nodal}$$

- Round trip time (RTT)

$$RTT = d_{c-s} + d_{s-c} + t_{server}$$

where t_{server} is server processing time

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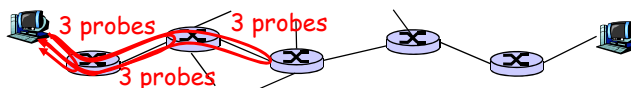
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"Real" Internet delays and routes

- What do "real" Internet delay & loss look like?
- **traceroute program**: provides delay measurement from source to router along end-end Internet path towards destination.

For all i :

- ❖ sends three packets that will reach router i on path towards destination
- ❖ router i will return packets to sender
- ❖ sender times interval between transmission and reply.



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"Real" Internet delays and routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr

Three delay measurements from
gaia.cs.umass.edu to cs-gw.umass.edu

```

1 cs-gw (128.119.240.254) 1 ms 1 ms 2 ms
2 border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145) 1 ms 1 ms 2 ms
3 cht-vbns.gw.umass.edu (128.119.3.130) 6 ms 5 ms 5 ms
4 jn1-at1-0-0-19.wor.vbns.net (204.147.132.129) 16 ms 11 ms 13 ms
5 jn1-so7-0-0-0.wae.vbns.net (204.147.136.136) 21 ms 18 ms 18 ms
6 abilene-vbns.abilene.ucaid.edu (198.32.11.9) 22 ms 18 ms 22 ms
7 nycm-wash.abilene.ucaid.edu (198.32.8.46) 22 ms 22 ms 22 ms
8 62.40.103.253 (62.40.103.253) 104 ms 109 ms 106 ms
9 de2-1.de1.de.geant.net (62.40.96.129) 109 ms 102 ms 104 ms
10 de.fr1.fr.geant.net (62.40.96.50) 113 ms 121 ms 114 ms
11 renater-gw.fr1.fr.geant.net (62.40.103.54) 112 ms 114 ms 112 ms
12 nio-n2.cssi.renater.fr (193.51.206.13) 111 ms 114 ms 116 ms
13 nice.cssi.renater.fr (195.220.98.102) 123 ms 125 ms 124 ms
14 r3t2-nice.cssi.renater.fr (195.220.98.110) 126 ms 126 ms 124 ms
15 eurecom-valbonne.r3t2.ft.net (193.48.50.54) 135 ms 128 ms 133 ms
16 194.214.211.25 (194.214.211.25) 126 ms 128 ms 126 ms
17 ***
18 ***
19 fantasia.eurecom.fr (193.55.113.142) 132 ms 128 ms 136 ms
  
```

trans-oceanic link

different packets

* means no response (probe lost, router not replying)

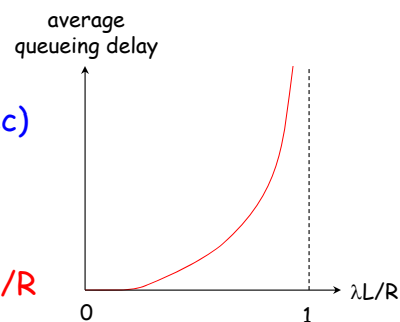
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Queueing delay (waiting time)

- R: link bandwidth (bps)
- L: packet length (bits)
 - ❖ service rate = R/L (pkts/sec)
- λ : average packet arrival rate

traffic intensity =
arrival rate/service rate = $\lambda L/R$



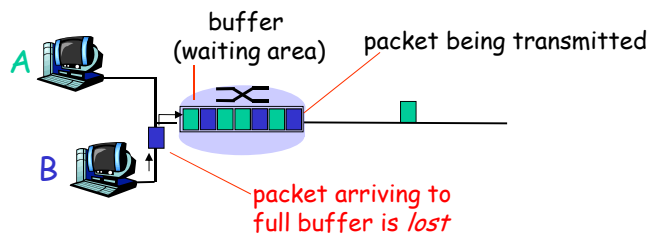
- $\lambda L/R \sim 0$: average queueing delay small
- $\lambda L/R \rightarrow 1$: delays become large
- $\lambda L/R > 1$: more "work" arriving than can be served, average delay infinite!
 - ❖ In reality, buffer overflow when $\lambda L/R \rightarrow 1$

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Introduction (SSL) 1-56

Packet loss

- buffer in router for each link has finite capacity
- lost packet may be retransmitted by previous node, by source end system, or not at all

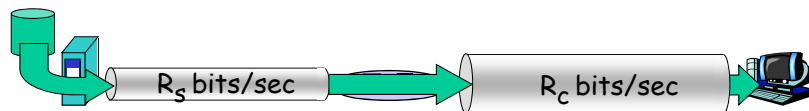


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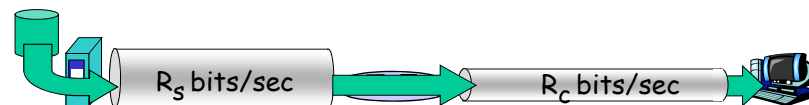
Introduction (SSL) 1-57

Throughput - rate at which bits are transferred from source to destination (in bits/sec.)

- $R_s < R_c$ end-end throughput less than ____ ?



- $R_s > R_c$ end-end throughput less than ____ ?



bottleneck link

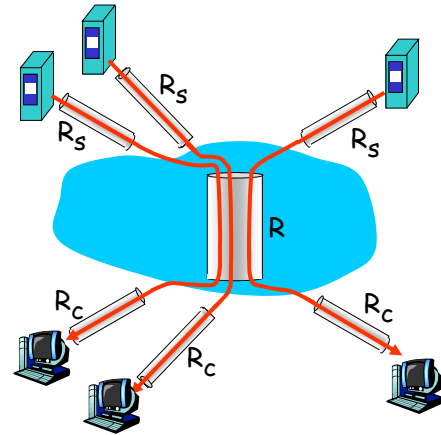
link on end-end path that constrains end-end throughput

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Introduction (SSL) 1-58

Throughput: Internet scenario

- per-connection end-to-end throughput is approximately $\min(R_c, R_s, R/10)$
 - ❖ *Actually sharing a bottleneck equally is ideal but unrealistic*
- In practice: R_c or R_s is often the bottleneck
- or the server is the bottleneck



10 connections (fairly) share backbone bottleneck link R bits/sec

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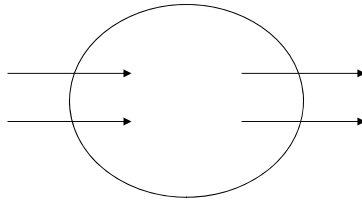
Little's law and a useful queueing delay formula

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Little's Law

Average population
= (average delay) x
(throughput rate)



$$\text{average delay} = \frac{1}{N} \sum_{i=1}^N \text{delay}_i$$

where N is number of departures

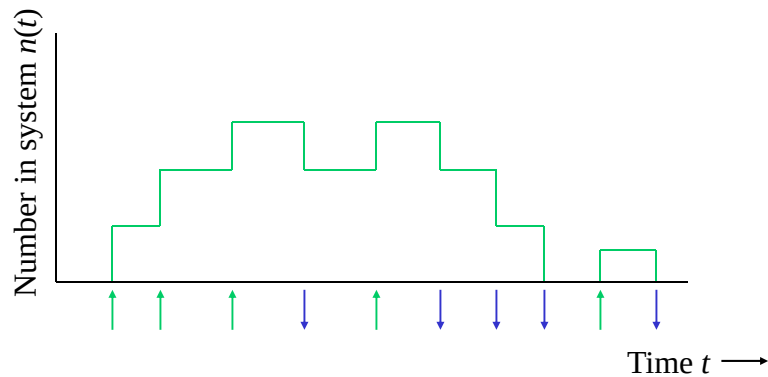
$$\text{throughput rate} = N/T$$

where T is duration of experiment

average population
(to be defined)

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Introduction (SSL) 1-61



$$\text{average population} = \frac{1}{\tau} \int_0^{\tau} n(t) dt$$

where τ is duration of the experiment

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Introduction (SSL) 1-62

random variable x

samples $x_1, x_2, \dots, x_n$

$$\text{mean (average)} \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\text{second moment} \quad \overline{x^2} = \frac{1}{n} \sum_{i=1}^n (x_i)^2 \geq (\bar{x})^2$$

$$\text{mean residual life} = \frac{\overline{x^2}}{2\bar{x}} \geq \frac{\bar{x}}{2}$$

Special case: x is a constant

$$\overline{x^2} = (\bar{x})^2$$

$$\text{mean residual life} = \frac{(\bar{x})^2}{2\bar{x}} = \frac{\bar{x}}{2}$$

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Introduction (SSL) 1-63

random variable x

with discrete values $x_1, x_2, \dots, x_m$

let p_i = probability $[x = x_i]$ for $i = 1, 2, \dots, m$

by definition

$$\text{mean} \quad \bar{x} = \sum_{i=1}^m x_i p_i$$

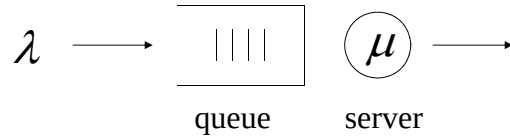
$$\text{second moment} \quad \overline{x^2} = \sum_{i=1}^m x_i^2 p_i$$

(Aside: For a continuous random variable,
use integration instead of summation.)

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Single-Server Queue



$\bar{x}$ average service time, in seconds

μ service rate, in jobs/second ($\mu = 1/\bar{x}$)

λ arrival rate, in jobs/second

ρ utilization of server

Conservation of flow

$$\lambda = \rho\mu$$

$$\rho = \frac{\lambda}{\mu} = \lambda\bar{x}$$

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Introduction (SSL) 1-65

M/G/1 queue

□ Single server

- ❖ does not idle when there is work, no overhead, i.e., it performs 1 second of work per second
- ❖ FIFO service

□ Arrivals according to a Poisson process at rate λ jobs/second

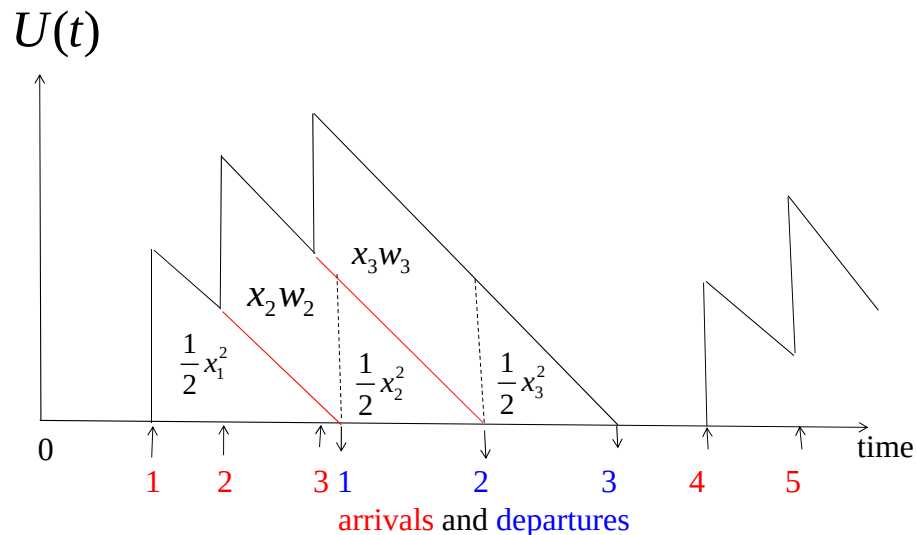
□ Service times of arrivals are $x_1, x_2, \dots, x_i \dots$ which are *independent, identically distributed* (with a **general** distribution)

□ Average service time is $\bar{x}$, average wait is W , average delay is $T = W + \bar{x}$

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Introduction (SSL) 1-66

Let $U(t)$ be the unfinished work at time t



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Introduction (SSL) 1-67

Derivation of W

Time average of unfinished work is

$$\begin{aligned}\bar{U} &= \frac{1}{\tau} \int_0^{\tau} U(t) dt \\ &= \frac{1}{\tau} \left(\frac{1}{2} \sum_{i=1}^n x_i^2 + \sum_{i=1}^n x_i w_i \right) \quad \text{\(x_i\) and \(w_i\) are independent} \\ &= \frac{n}{\tau} \left(\frac{1}{2} \overline{x_i^2} + \overline{x_i} \times \overline{w_i} \right) \quad \text{where } \overline{x_i w_i} = \overline{x_i} \times \overline{w_i}\end{aligned}$$

For Poisson arrivals, the average wait is equal to $\bar{U}$ from the *Poisson arrivals see time average (PASTA) Theorem*

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Derivation of W (cont.)

□ The average wait is

$$W = \lambda \left(\frac{1}{2} \overline{x^2} + \bar{x}W \right) = \frac{\lambda \overline{x^2}}{2} + \lambda \bar{x}W$$

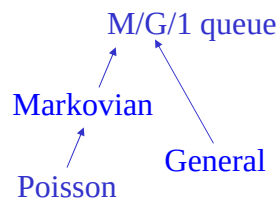
$$W(1 - \rho) = \frac{\lambda \overline{x^2}}{2}$$

$$W = \frac{\lambda \overline{x^2}}{2(1 - \rho)}$$

Pollaczek-Khinchin (P-K)
mean value formula

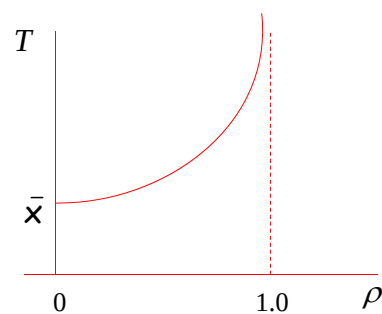
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Average delay is

$$T = \bar{x} + W = \bar{x} + \frac{\lambda \overline{x^2}}{2(1 - \rho)}$$



Also called Pollaczek-Khinchin (P-K) mean value formula

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Special Cases

1. Service times have an exponential distribution (M/M/1). We then have

$$\overline{x^2} = 2(\overline{x})^2$$

$$W = \frac{\lambda(2)(\overline{x})^2}{2(1-\rho)} = \frac{\lambda(\overline{x})^2}{1-\rho} = \frac{\rho(\overline{x})}{1-\rho}$$

$$T = W + \overline{x}$$

$$= \frac{\rho\overline{x}}{1-\rho} + \overline{x} = \frac{\rho\overline{x} + \overline{x} - \rho\overline{x}}{1-\rho}$$

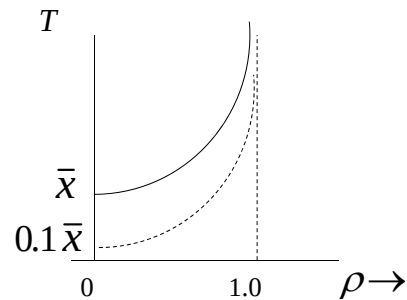
$$= \frac{\overline{x}}{1-\rho} = \boxed{\frac{\rho}{1-\rho} \frac{1}{\lambda}}$$

T decreases as λ increases

$$\lambda \rightarrow 10\lambda$$

$$\mu \rightarrow 10\mu$$

$$\rho = \frac{10\lambda}{10\mu} = \frac{\lambda}{\mu}$$



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Introduction (SSL) 1-71

2. Service times are constant (deterministic)

$$\overline{x^2} = (\overline{x})^2$$

↓
M/D/1

$$W = \frac{\lambda(\overline{x})^2}{2(1-\rho)} = \frac{\rho\overline{x}}{2(1-\rho)}$$

$$T = W + \overline{x}$$

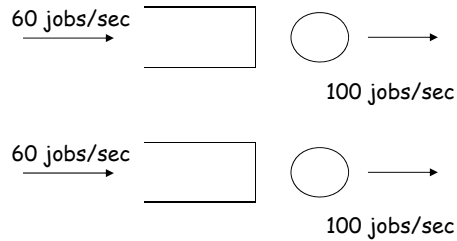
$$\boxed{T = \frac{\rho(2-\rho)}{2(1-\rho)} \frac{1}{\lambda}}$$

T decreases as λ increases

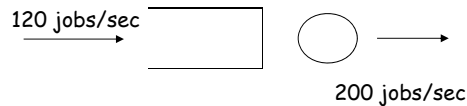
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Two Servers and Two Queues:



Single Higher Speed Server:



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Chapter 1: roadmap

- 1.1 What *is* the Internet?
- 1.2 Network edge
 - end systems, access networks, links
- 1.3 Network core
 - circuit switching, packet switching, network structure
- 1.4 Delay, loss and throughput in packet-switched networks
- 1.5 Protocol layers, service models
- 1.6 Networks under attack: security
- 1.7 History

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Protocol "Layers"

Networks are complex!

- ❑ many "pieces":
 - ❖ hosts
 - ❖ routers
 - ❖ links of various media
 - ❖ applications
 - ❖ protocols
 - ❖ hardware, software

Question:

How to organize network structure?

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Layered architecture

- ❖ Use abstraction to **hide complexity**
- ❖ Each layer
 - provides a service via its own internal actions as well as relying on service provided by layer below
 - is a **network of processes**
- ❖ Can have alternative abstractions at each layer (resulting in **protocol graph** rather than **protocol stack**)

Application programs
Process-to-process channels
Host-to-host connectivity
Hardware

Application programs	
Request/reply channel	Message stream channel
Host-to-host connectivity	
Hardware	

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Why layering?

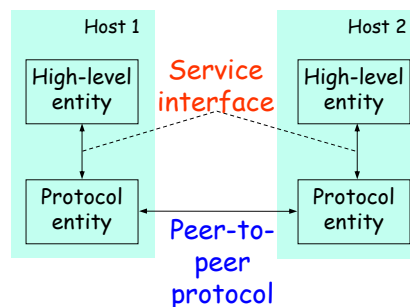
- ❖ layered architecture as **reference model** for protocol design by community effort
 - decompose a large system into smaller pieces which can be designed and implemented by different people/teams
- ❖ **modularity** eases maintenance and evolution of system
 - allows changes in implementation method so long as API remains the same, e.g., different Ethernets
- ❖ strict layering **often violated** for efficient protocol implementation
 - cross-layer design

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Each protocol

- ❖ involves two or more peers
- ❖ two kinds of specifications
 - **service interface**: operations a local user can perform on a protocol entity and get results
 - **peer-peer protocol**: format and meaning of messages exchanged by protocol entities (also called **peers**) to provide protocol service
- ❖ The term "protocol" generally refers to peer-peer spec

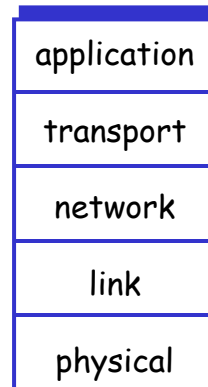


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Internet protocol stack

- ❑ **application:** supporting network applications
 - ❖ FTP, SMTP, HTTP
- ❑ **transport:** process-process data transfer
 - ❖ TCP, UDP
- ❑ **network:** routing of datagrams from source to destination
 - ❖ IP, routing protocols
- ❑ **link:** data transfer between neighboring network elements
 - ❖ PPP, Ethernet, 802.11 (WiFi)
- ❑ **physical:** bits "on the wire"

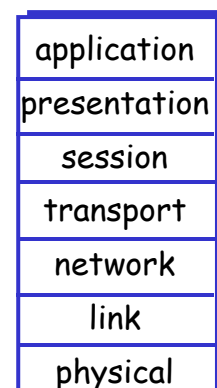


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ISO/OSI reference model

- ❑ **presentation:** allow applications to interpret meaning of data, e.g., encryption, compression, machine-specific conventions
- ❑ **session:** synchronization, checkpointing, recovery of data exchanged
- ❑ Internet stack "missing" these layers!
 - ❖ *these services, if needed, must be implemented in application (or application protocol)*
 - ❖ *needed?*

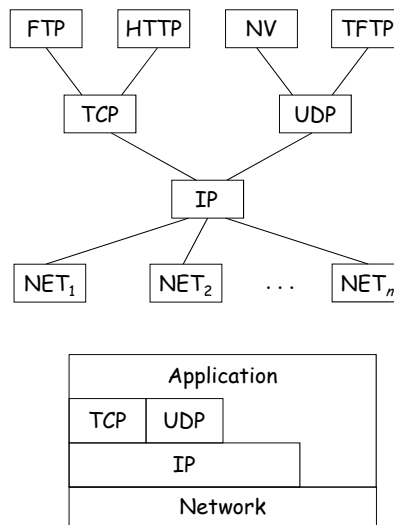


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Internet Architecture

- ❑ Internet Engineering Task Force (IETF)
- ❑ application protocols support applications
- ❑ multiplexing and demultiplexing
- ❑ *hourglass* shape (only IP in network layer)
 - ❖ best effort service
=> any delivery service can be used by IP
- ❑ limitation of hourglass

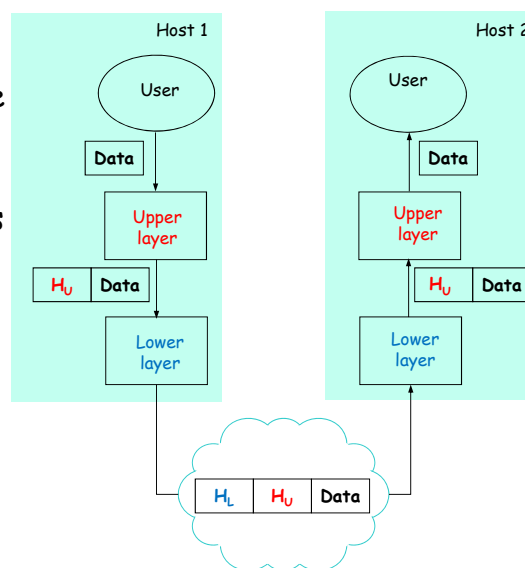


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Encapsulation

- ❑ Protocol peers provide a data delivery service
- ❑ How do protocol peers in different machines exchange protocol messages between themselves?
 - ❖ In **protocol header** encapsulated and de-encapsulated



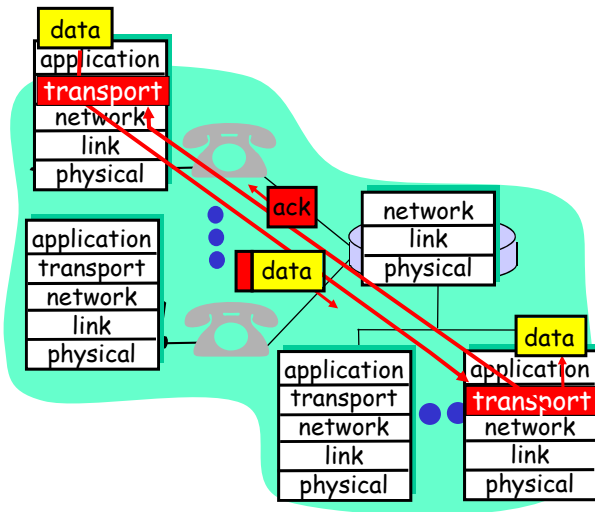
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Logical communication between peers

E.g.: transport

- accept data from application
- add addressing, reliability check info to form a message
- send message to peer via a delivery service
- wait for peer's reply (ack)



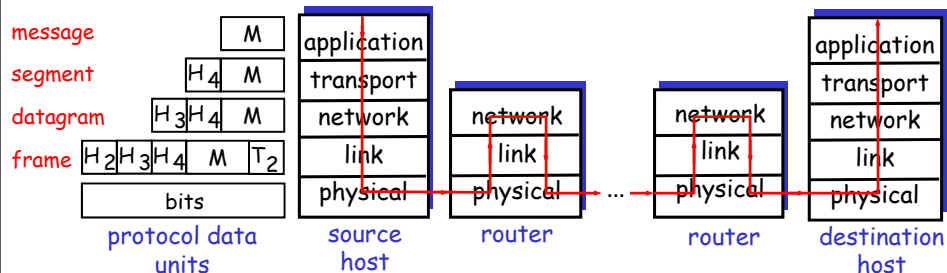
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Physical path of data

Each layer takes data (**service data unit**) from above

- adds header to create its own **protocol data unit**
- passes protocol data unit to layer below



Note: In the past, a **switch** implements only two layers (physical and link). Nowadays many switches function as routers

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Chapter 1: roadmap

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- 1.7 History

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Introduction (SSL) 1-85

Network Security

- The field of network security is about:
 - ❖ how bad guys can attack computer networks
 - ❖ how we can defend hosts and networks against attacks
 - ❖ how to design architectures that are immune to attacks
- Internet not originally designed with (much) security in mind
 - ❖ *original vision*: "a group of mutually trusting users attached to a transparent network" ☺
 - ❖ needs security considerations in each layer

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Bad guys: put malware into hosts via Internet

- ❑ malware can get in host from:
 - ❖ *virus*: self-replicating infection from receiving/executing object (e.g., e-mail attachment)
 - ❖ *worm*: self-replicating infection from passively receiving object that gets itself executed
- ❑ *spyware* can record keystrokes, web sites visited, upload info to collection site
- ❑ infected host can be enrolled in a *botnet*, used for spam, DDoS attacks

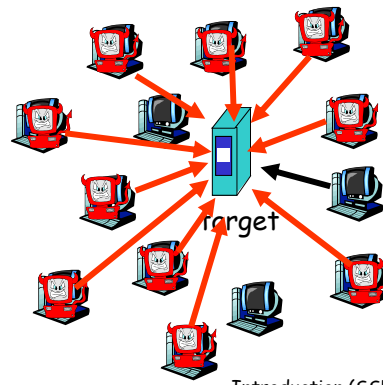
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Distributed Denial of service (DDoS) attacks

- ❑ attackers overwhelm resources with bogus traffic
 - ❖ make resources (server, bandwidth) unavailable to legitimate traffic

1. select target
2. break into hosts around the network (see botnet)
3. send packets toward target from compromised hosts



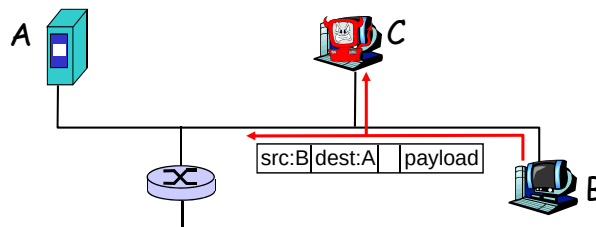
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The bad guys can sniff packets

Packet sniffing:

- ❖ broadcast media (shared Ethernet, wireless)
- ❖ promiscuous network interface reads/records all packets (e.g., including passwords!) passing by



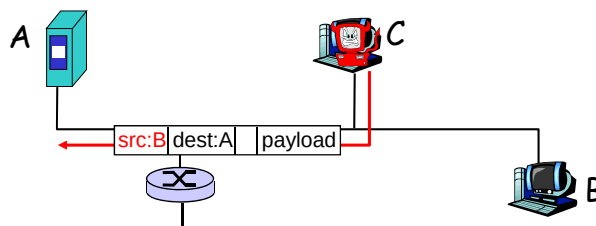
- ❖ Wireshark software is a (free) packet-sniffer

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The bad guys can use false source addresses

- ❑ *IP spoofing:* send packet with false source address



More on security in Chapter 8

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End of Chapter 1

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